



Postgres-XC

A PostgreSQL Clustering Solution

What started with a simple relational database system, is expanding its horizons by developing new technology that satiates the ever-increasing need for more data storage, greater transaction throughput and higher availability. Using a cluster to solve these scalability problems is a present trend. This article talks about Postgres-XC, a clustering solution based on the popular PostgreSQL RDBMS.

A cluster is a collection of commodity components that provide scalability and availability at a low cost to the consumer. A database cluster is a collection of database servers that store and process data using commodity hardware, satisfying the need for more data storage, higher throughput and providing high availability. Postgres-XC is such a database cluster system; it is based on the world's most advanced open source database, PostgreSQL, and follows the same open source model.

The Postgres-XC project began in 2009, through a collaboration between NTT and EnterpriseDB. The goal was to build an open source clustering solution based on PostgreSQL with 100 per cent compatible client APIs. Having PostgreSQL-compatible APIs allows existing PostgreSQL applications to use Postgres-XC with little (or no) change. The licensing terms of this project are the same as that of PostgreSQL.

Postgres-XC architecture

Postgres-XC is a write-scalable, synchronous multi-master, transparent PostgreSQL clustering solution based on shared-

nothing architecture. It is a collection of tightly coupled database components, which can be installed on one or more physical or virtual machines. The components do not share any resources such as disk, cache or memory.

- *Write-scalability* means that Postgres-XC can be configured with as many database servers as needed; Postgres-XC is able to handle more writes than a single PostgreSQL server.
- *Multi-master* implies that clients can connect to multiple database servers, and that each database server provides a single, consistent, cluster-wide view of the database.
- *Synchronous* means that a write from any of the masters is immediately visible to other transactions running on other masters.
- *Transparent* means that applications do not have to worry about how the data is stored in multiple database servers, internally.

Figure 1 gives the architectural overview of Postgres-XC with its three main components:

1. *Global Transaction Manager (GTM)* gathers and